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B

work done by external force $= \Delta U = q\Delta V$

The net potential at a point is the scalar sum of the potentials due to the first two point charges. ΔU is independent of the path.

Since the charges have opposite polarity,

- the electric potential is positive near $+2q$ and negative near $-q$.
- the point where the electric potential is zero is between the two charges, and nearer to $-q$.

A: always equidistant from $+2q$, starts and ends about the same distance from $-q$. $\Delta V \approx 0$

C: starts and ends about equidistant from both $+2q$ and $-q$. $\Delta V \approx 0$

D: $V_{initial}$ is a large negative number, and V_{final} is a smaller negative number. ΔV is positive.

B: $V_{initial}$ is a large negative number, and V_{final} is a large positive number. ΔV is the greatest.

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A

Resistance R is the ratio of V to I